COMP 551 Assignment 3 Report

Hadi El Nawfal, Bora Sanal, Andrea Gómez

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1 Abstract

This work addresses the problem of visual outlier detection in groups of five 32×32 grayscale images, where four images share a latent common property and one does not. The task is formulated as a 5-class classification problem over 3,000 training groups, with a random-chance baseline of 20%. The principal challenges are learning inter-image relational similarity under a strict constraint of 25,000 trainable parameters and maintaining robustness to the label noise present in the dataset. We propose a neural architecture that combines a parameter-efficient convolutional encoder built with depthwise separable convolutions. We also use squeeze-and-excitation blocks, a cross-image self-attention module with diagonal masking, and a pairwise relation network that explicitly compares each candidate to every other group member. Relation vectors are aggregated using attention-weighted averaging, max pooling, and standard deviation, and are further enriched with prototype-based scalar similarity features before being passed to a scoring head. Images are preprocessed through adaptive foreground cropping and centering before global standardization. Training augmentation includes group-consistent geometric transformations, position permutation, shared random erasing, and shared Gaussian noise. To maximize generalization, we train a multi-seed ensemble of 30 models (five 5-fold cross-validation runs with different seeds plus five full-data models), apply stochastic weight averaging, and combine predictions through squared-accuracy-weighted logit averaging with 8-variant test-time augmentation. The final model uses 24,997 trainable parameters and achieves a public Kaggle leaderboard accuracy of 76.8%.

2 Introduction

Standard image classification tasks typically involve assigning a label to a single image based on its visual content. The task considered in this project differs fundamentally: given a group of five grayscale images, the model must determine which one does not belong. Four of the images share a latent visual property, and the goal is to identify the index of the outlier. Since this shared property is never explicitly defined, the model cannot rely on recognizing objects or attributes in isolation. Instead, it must learn to compare images within a group and reason about their relative similarities. Several factors make this problem particularly challenging. The training set consists of only 3,000 groups, which severely limits the data available for learning complex visual relationships. At the same time, the model is constrained to at most 25,000 trainable parameters, requiring careful architectural decisions to balance expressiveness with efficiency. The dataset also contains noisy labels, meaning the model must tolerate occasional incorrect supervision without overfitting to those errors. Our approach is built around the idea that outlier detection in this setting requires both local pairwise comparisons and a global understanding of group structure. We design a network that encodes each image into a shared embedding space using a parameter-efficient convolutional encoder, then refines these representations through a cross-image self-attention mechanism with diagonal masking that allows each image to attend only to the other group members. The model then performs explicit pairwise comparisons through a learned relation network, aggregates the results using attention-derived weights alongside prototype-based similarity statistics, and scores each position accordingly. To compensate for the small dataset, we train a multi-seed ensemble of 30 models across different data splits and initializations, apply stochastic weight averaging, and combine predictions through squared-accuracy-weighted logit averaging with 8-variant test-time augmentation.

3 Data Preprocessing and Exploration

The dataset consists of 3,000 training groups, each containing five 32×32 grayscale images with pixel values in $[0, 255]$. We first rescale all pixels to $[0, 1]$ and then apply a foreground extraction step: for each image, pixels below an adaptive threshold are treated as background, a tight bounding box is computed around the foreground, and the cropped region is resized to fit within a 26×26

area while preserving aspect ratio before being centered on a blank 32×32 canvas. This removes variation in object position and scale, allowing the model to focus on visual content. Global mean and standard deviation are then computed from the training set and used to standardize both training and test images.

A key insight that shaped our augmentation strategy is that the outlier signal is inherently relational, existing in the differences between images rather than in any single image. In earlier iterations, per-image transforms such as independent random erasing, additive noise, and cross-group mixup introduced artificial differences within groups, corrupting the outlier signal. Identifying and removing these issues was one of the most impactful changes during development. Our final augmentation pipeline operates strictly at the group level: random permutation of image order with label update (50%) to encourage position invariance, group-consistent geometric transforms (horizontal flip, vertical flip, and 90° rotations applied identically to all five images), shared random erasing where the same small patch is removed from all images simultaneously (15%), and shared Gaussian noise where a single noise field is added to all five images (10%, $\sigma = 0.02$). In every case, because the same transform is applied across the entire group, the relative differences between images are preserved.

For validation, we use 5-fold stratified cross-validation repeated with five different random seeds (42, 7, 123, 17, 99), producing 25 fold models. Stratification preserves label distribution across folds, and multiple seeds introduce diversity in both fold partitions and weight initialization, reducing the risk that ensemble performance depends on a single favorable split.

4 Methods

The model architecture consists of two main components: a convolutional encoder that extracts feature embeddings from each image, and a relational scoring module that identifies the outlier within the group. Throughout development, we iterated through several architectural variants before arriving at the final design. Below we describe each component in detail, followed by the optimization setup and ensemble strategy.

4.1 Image Encoder

Each 32×32 grayscale image is processed independently by a shared convolutional encoder with identical weights across all five group members. The full layer configuration is shown in Table 1. The encoder begins with a 5×5 convolution (without bias) that expands the single input channel to 16 feature maps, followed by batch normalization, ReLU, and 2×2 max pooling. The subsequent three blocks use depthwise separable convolutions, each consisting of a per-channel 3×3 spatial filter followed by a 1×1 pointwise projection, both with batch normalization and ReLU. This factorization substantially reduces parameter count compared to standard convolutions while maintaining representational capacity, which is critical under the 25,000 parameter budget. The channel dimensions progress as 16→28→40→48, with 2×2 max pooling reducing spatial resolution from 32→16→8→4 after the first three stages. A squeeze-and-excitation block follows the final convolutional block, computing a channel-wise attention vector through a two-layer bottleneck (48→8→48 with ReLU and sigmoid) that rescales each channel according to its learned importance. Adaptive average pooling then collapses spatial dimensions to 1×1, and a final linear layer projects the resulting 48-dimensional vector to a 56-dimensional embedding. Because the same encoder processes all five images with shared weights, the model learns consistent representations while storing only a single copy of the encoder parameters.

Layer	**Out Ch.**	**Kernel**	**Other**
Conv2D	16	5×5	pad=2
BN + ReLU	16	–	–
MaxPool	16	2×2	$32 \to 16$
DWSep 2	28	$3 \times 3 + 1 \times 1$	dw + pw
MaxPool	28	2×2	$16 \to 8$
DWSep 3	40	$3 \times 3 + 1 \times 1$	dw + pw
MaxPool	40	2×2	$8 \to 4$
DWSep 4	48	$3 \times 3 + 1 \times 1$	dw + pw
SE Block	48	–	$48 \to 8 \to 48$
AAP	48	–	1×1
Linear	56	–	$48 \to d$

Table 1: Shared convolutional encoder architecture.

4.2 Architecture Variants

Before arriving at the final architecture, we experimented with several variants that differed in how they compared the embeddings of the five images. In all cases, each image is first processed by the shared convolutional encoder to produce an embedding vector z_i.

OddOneOutNet. The first architecture compares each image embedding to the mean embedding of the other four images in the group. For a candidate image i, the model computes the average representation of the remaining images and measures how different the candidate is from this group mean. These features are

passed to a small multilayer perceptron that produces an outlier score for each position. While simple and parameter-efficient, this approach relies heavily on the mean representation and may miss more subtle pairwise inconsistencies.

PairwiseOddOneOutNet. To better capture relationships between images, the second architecture introduces explicit pairwise comparisons. For each candidate image, the model computes the difference between its embedding and the embeddings of the other four images. These differences are aggregated into a single feature vector that summarizes how dissimilar the candidate is from each group member individually. The aggregated representation is then passed to a scoring network to determine the likelihood that the image is the outlier.

StrongScorerOddOneOutNet. In this variant, we extended the scoring network by providing it with richer relational features. For each candidate image, the model constructs a feature vector that includes the mean embedding of the other images, the average and maximum pairwise differences, and the absolute difference between the candidate embedding and the group mean. These features provide both global and local information about how each image relates to the group.

RelationNetOddOneOut. We also experimented with a relation-network-based architecture that explicitly learns pairwise relationships through a dedicated neural network. For each pair of images, a small MLP processes a combination of their embeddings, their element-wise difference, and their element-wise product. These learned relation vectors are aggregated via mean and max pooling. This approach allows the model to learn which types of differences are most indicative of an outlier, rather than relying on handcrafted distance metrics.

4.3 Final Architecture

The final model extends the relation network approach with three key additions: cross-image self-attention, attention-guided aggregation of relation vectors, and prototype-based scalar similarity features.

All embeddings are first ℓ_2-normalized to place them on the unit hypersphere, stabilizing subsequent similarity computations. A single-head scaled dot-product self-attention module then operates across all five image embeddings. The attention uses a key dimension of 16 with a diagonal mask that prevents each image from attending to itself, forcing the attention to capture only inter-image relationships. Attention dropout (0.05) is applied for regularization. The output is combined through a linear projection with a residual connection and layer normalization, allowing each representation to be refined in the context of the other group members before any pairwise comparison takes place.

For each candidate position i, the model performs explicit pairwise comparisons with every other image $j \neq i$. Each pair produces a feature vector

$$[z_i, z_j, |z_i - z_j|, z_i \cdot z_j]$$

of dimension $4 \times 56 = 224$, which is processed by a two-layer relation MLP ($224 \to 40 \to 40$ with ReLU and dropout). The four resulting relation vectors are then aggregated in three complementary ways: an attention-weighted mean where the weights are derived from the attention scores of image i to each other image (re-normalized to sum to 1), element-wise max pooling, and element-wise standard deviation. These three 40-dimensional vectors capture different aspects of the candidate's relationship to the group: the attention-weighted mean emphasizes the comparisons the model deems most relevant, the max captures the strongest single discrepancy, and the standard deviation measures how consistently the candidate differs from the other images.

This 120-dimensional aggregation is further enriched with four prototype-based scalar features: cosine similarity between the candidate and the prototype (mean of the other four embeddings), mean squared ℓ_2 distance to the prototype, mean cosine similarity to all other images individually, and the standard deviation of these cosine similarities. The standard deviation is particularly informative, as a true outlier tends to have uniformly low similarity to all group members, yielding low variance. The resulting 124-dimensional vector is passed through a two-layer scoring MLP ($124 \to 16 \to 1$ with ReLU and dropout) that outputs a single logit per position. The five logits are concatenated to form the 5-way classification output.

4.4 Loss Function and Optimization

Training is formulated as a 5-class classification problem using cross-entropy loss with label smoothing of 0.08 to mitigate the effect of noisy labels in the dataset. The model is optimized with AdamW using a learning rate of 2.0×10^{-3} and a weight decay of 4×10^{-4}.

For the first 65% of training, a OneCycleLR schedule is used with cosine annealing, a 25% warmup phase, a peak-to-initial learning rate ratio of 15, and a final learning rate 300× lower than the peak. After 65% of epochs, the optimizer transitions to stochastic weight averaging (SWA) with a fixed learning rate of 3×10^{-4} and an annealing period of 8 epochs. SWA averages model weights over the final phase of training, producing flatter minima that generalize better.

At the end of training, batch normalization statistics are recomputed for the SWA model, and the final model is selected as whichever achieves higher validation accuracy between the SWA average and the best single checkpoint. Gradients are clipped to a maximum norm of 1.0, and training uses a batch size of 64 with early stopping (patience of 40 epochs, maximum 150 epochs per fold).

4.5 Ensemble and Inference Strategy

To maximize generalization from the limited training set, we train a multi-seed ensemble. We run 5-fold stratified cross-validation five times with different random seeds (42, 7, 123, 17, 99), producing 25 fold models. Each seed yields different fold partitions and weight initializations, ensuring diversity across ensemble members. In addition, five full-data models are trained on all 3,000 groups, one per seed, for 160 epochs using cosine annealing with SWA starting at epoch 100. This yields a total of 30 models.

At inference, logits from all 30 models are combined using weights proportional to the square of each model's validation accuracy, which sharpens contributions toward stronger models.

We further apply 8-variant test-time augmentation corresponding to the complete D_4 dihedral symmetry group of the square (four 90° rotations each combined with a horizontal flip). Logits are averaged across all $30 \times 8 = 240$ forward passes before taking the arg max. This TTA strategy is consistent with our group-level augmentation design, as every variant preserves relative differences between images.

4.6 Parameter Constraint

The final model contains 24,997 trainable parameters, utilizing 99.99% of the 25,000 budget. The parameter breakdown is: encoder 8,644 (depthwise separable convolutions + SE block), attention 3,696 (single head with key dimension 16), relation MLP 10,640, and scorer 2,017.

Throughout development, embedding dimension (56), channel counts ($16 \to 28 \to 40 \to 48$), attention head dimension (16), relation hidden size (40), scorer hidden size (16), and dropout rates (0.10 model, 0.05 attention) were tuned jointly to maximize representational capacity while remaining strictly under the limit. The use of depthwise separable convolutions was instrumental in fitting a deeper encoder within the budget. All 30 models in the ensemble share the same architecture.

Component	**Parameters**	**(% of Total)**
Encoder (DW + SE)	8,644	34.6%
Self-Attention	3,696	14.8%
Relation MLP	10,640	42.6%
Scorer MLP	2,017	8.1%
Total	**24,997**	**100%**

Table 2: Parameter breakdown of the final model (25,000 parameter budget).

Configuration	**Public Acc. (%)**	**Δ**
25 folds, uniform weights, no TTA	76.00	–
25 folds, uniform weights, 8-TTA	76.00	+0.00
25 folds, squared weights, 8-TTA	76.00	+0.00
All 30 models, squared weights, 8-TTA	76.80	+0.80

Table 3: Ablation of ensemble composition, weighting, and test-time augmentation (TTA) on public test accuracy.

5 Results

5.1 Internal Validation

We evaluate model performance through 5-fold stratified cross-validation repeated across five random seeds, yielding 25 independently trained fold models. The mean fold validation accuracy across all 25 models is 90.83%, with values ranging from 88.67% to 93.33%. The relatively low variance across seeds confirms that performance is consistent under different data splits and initializations. Notably, the best checkpoint outperformed SWA in all 25 folds, indicating that with 150 epochs and a patience of 40, the early stopping mechanism consistently identified strong local minima that SWA could not further improve.

5.2 Ablation Study

To isolate the contribution of each inference-time component, we evaluate four configurations on the public test set.

An interesting finding is that neither TTA nor squared-accuracy weighting improved accuracy over the 25-fold uniform ensemble on the public test set. This suggests that with 25 diverse fold models already averaging out prediction noise, the additional variance reduction from TTA becomes redundant. The weighting strategy likely has minimal effect because fold validation accuracies are tightly clustered (88.67%–93.33%), leaving little room for squared weighting to differentiate between models. The only measurable gain (+0.80 points) comes from adding the five full-data models, which benefit from training on all 3,000 groups rather than the 2,400 avail-

able to each fold model. This indicates that increased data exposure, rather than inference-time techniques, is the primary limiting factor for this ensemble.

5.3 Architecture Evolution

Over the course of development, we iterated through multiple architectural variants, progressively improving from the logistic regression baseline (19.4%) to the final public test accuracy of 76.80%. The most impactful architectural transitions include the shift from mean-based to pairwise comparison, the introduction of depthwise separable convolutions and squeeze-and-excitation blocks to enable a deeper encoder within the parameter budget, and the addition of cross-image self-attention.

In the final phase, the architecture was fixed at 24,997 parameters and all gains came from training improvements: expanding from 3 to 5 seeds ($18 \rightarrow 30$ models), earlier SWA activation (65% instead of 75%), and increased label smoothing (0.08 instead of 0.05). The gap between the mean fold validation accuracy (90.83%) and the public test accuracy (76.80%) suggests a meaningful distribution shift between the training and test sets, reinforcing the importance of ensemble diversity and robust training strategies.

5.4 Public Kaggle Score

The final public leaderboard accuracy is 76.80%, obtained using all 30 models with squared-accuracy-weighted logit averaging and 8-variant TTA ($30 \times 8 = 240$ forward passes per test sample). The model uses 24,997 of the allowed 25,000 trainable parameters. This represents an increase from the random-chance baseline of 20% to 76.8% in our final model.

6 Discussion and Conclusion

The final architecture achieves a high level of expressiveness while remaining under the parameter constraint. The use of depthwise separable convolutions connected to a squeeze-and-excitation network improves feature efficiency. This allows to get richer representations while staying under the parameter budget.

Additionally, the decision to use self-attention with diagonal masking both forces inter-image reasoning instead of independent encoding, as it's taking the attention away from specific images. This way, the model focuses on relational differences, which comes down to being the core problem of this task.

The aggregation combines mean, max, and standard deviation of relation vectors to capture the global consensus, strongest discrepancies, and signal consistency respectively. These are enriched with four prototype-based scalar features: cosine similarity to the group mean (alignment with group representation), mean squared L2 distance to the group mean (deviation from group center), average cosine similarity to other images (overall group similarity), and standard deviation of these similarities (whether the image is uniformly dissimilar to all members or only to a subset). Together, these statistics make outlier detection more stable and interpretable. The group-consistent augmentation strategy further strengthens generalization by preserving relational signals across all transforms.

The ensemble of 30 models, plus the Stochastic Weight Average and Test-time augmentation boost performance by reducing variance from the small dataset, mitigating label noise and showing strong gains from diversity across folds and seeds.

Through this architecture, we demonstrate that carefully designed relational inductive biases can outperform larger generic models. Through the model parameter constraint, we demonstrate that efficiency doesn't necessarily limit performance, and that smart design can compensate for model size. Through explicit relational modeling, this is, combining learned pairwise comparisons, a group-level reference representation, and cross-image contextualization mechanisms, we strenghten the notion that the model should be designed for comparison, not just recognition. For this case, a pure CNN-based approach would be insufficient for this task.

Additionally, we make a point in the fact that the training strategy (augmentation, ensembling, Stochastic Weight Averaging and Test-Time Augmentation) is as important as the architecture. This shows how important optimization and generalization can be for low-data regimes.

The fall in test accuracy with respect to the validation results indicates that there still are remaining challenges in robust generalization, and that the model is sensible to dataset noise and unseen patterns. Future improvements could include better regularization against overfitting; more robust attention mechanisms, such as multi-head attention; improved handling of label noise; or exploring contrastive or metric learning objectives. Overall, this project provides a strong demonstration that structured relational reasoning and efficient deep learning design can effectively tackle complex visual comparison tasks.

7 Statement of Contributions

Andrea Gómez was in charge of building the model architecture, for which she iterated through slightly different model variants, and wrote the discussion. Hadi

El Nawfal focused on enhancing the model architecture and carried out extensive hyperparameter tuning, implementing and validating improvements to achieve more reliable and optimized results. Bora Sanal contributed by conducting in-depth technical research to identify strategies for improving the model's performance and supported the team in structuring and refining the report with these insights.

8 Statement on the Use of LLMs and Other Resources

Large Language Model (ChatGPT) and AI-based writing assistant (Grammarly) were used to improve the clarity, grammar, and academic tone of the written content. ChatGPT was also consulted to help identify, iterate and fix bugs during the code development process. No code or data was shared with other teams.